Project Synopsis: Fraud Shield – Real-Time Anomaly Detection for Digital Micro-Transactions

A comprehensive academic project synopsis engineered for B.Tech Artificial Intelligence and Machine Learning (AIML). Fully detailed and ready for submission.

Project title: Fraud Shield – Real-Time Anomaly Detection for Digital Micro-Transactions

Team name & ID: Quantumm Coders , Team ID = 40

Institute / Course: Gla university, B.Tech CSE AI/ML

Version: v1.0

Date: 09 Oct 2026

GitHub repository link- https://github.com/redhacker31/gla_mini_project1

1. Overview

Problem statement: Vulnerable first-time digital wallet users in rural India face increasing low-value transaction fraud (such as velocity bursts, collect-request phishing, smurfing, and micro-draining). Traditional static monetary thresholds fail to detect these attacks because individual amounts fall beneath conventional alarm limits (<INR 500), while mature supervised models suffer from acute cold-start failures due to minimal historical data.

Goal: Build an edge-compatible, real-time AI/ML behavioral anomaly detection and adaptive risk-based authentication system ('Fraud Shield') that inspects micro-transactions within a sub-50ms inference budget, isolates anomalous velocity patterns via unsupervised learning (Isolation Forest), and selectively challenges suspicious transactions with Step-Up Multi-Factor Authentication (SMS OTP verification).

Non-goals: Direct core banking switch interception without authorized sandbox APIs, credit underwriting, physical biometric hardware integration, or nationwide commercial settlement operations in v1.

Value proposition: Sub-50ms behavioral risk scoring, cold-start resilience using population-level priors, frictionless pass-through for legitimate routine commerce, and dynamic Step-Up verification that blocks unauthorized micro-siphoning while minimizing user verification fatigue.

2. Scope and Control

2.1 In-scope

Ingestion and schema validation of streaming micro-transaction payloads via asynchronous REST API; backward-looking temporal velocity feature engineering (5m, 1h, 24h windows, inter-arrival time deltas, recipient entropy, amount deviation ratios); unsupervised anomaly detection via Isolation Forest paired with deterministic security rule blending into a calibrated Composite Risk Score S(x) in [0.0, 1.0]; three-tiered risk decision engine (Low Risk: Allow, Medium Risk: Step-Up OTP challenge, High Risk: Temporary freeze / security review); secure Step-Up OTP service (CSPRNG 6-digit tokens, SHA-256 hashing, 180s

TTL, max 3 attempts rate limiter, mock/Twilio SMS integration); real-time monitoring dashboard displaying throughput, anomaly volume, dispute rates, and latency telemetry; comprehensive empirical evaluation on PaySim mobile money and ULB credit card fraud benchmark datasets.

2.2 Out-of-scope

Live direct interception of commercial banking core switches without authorized sandbox credentials; storage of unmasked personal identifiers, Aadhaar numbers, or permanent bank credentials; native compiled iOS/Android binary apps in v1 (responsive web dashboard & REST API provided); complex multi-hop graph neural networks requiring multi-second GPU cluster compute in v1.

2.3 Assumptions

- Micro-transactions are evaluated as discrete real-time events via an authorized payment gateway webhook or client application simulator.
- Each transacting account possesses a registered mobile number capable of receiving SMS out-of-band tokens.
- The evaluation relies on standardized benchmark transaction logs (PaySim, ULB Credit Card) exhibiting realistic extreme class imbalance (<0.2% fraud).
- Transaction payloads contain standard metadata (user ID, timestamp, amount, recipient VPA/account, device/channel indicators).

2.4 Constraints

Strict real-time end-to-end API response budget: inference latency must remain <50ms (p95) to satisfy payment gateway SLAs; academic timeline: 16-week B.Tech capstone execution window; computational budget: standard commodity workstations / cloud instances without requiring dedicated high-cost GPU clusters; regulatory constraints: strict adherence to India's Digital Personal Data Protection (DPDP) Act 2023 (data minimization, pseudonymization) and RBI Master Directions on Digital Payment Security Controls.

2.5 Dependencies

Python 3.11+ ecosystem (FastAPI, scikit-learn, XGBoost, pandas, NumPy); relational database persistence (SQLite for local dev/testing; PostgreSQL for production); telecom SMS gateway API or mock SMS simulator for out-of-band OTP delivery; public benchmark repositories (PaySim Kaggle repository, ULB Credit Card Machine Learning Group).

2.6 Acceptance criteria and sign-off

- GIVEN a legitimate micro-transaction with normal user velocity WHEN evaluated by the risk engine THEN it receives Composite Score S(x) < 0.45 and is approved in <50ms without OTP prompt.
- GIVEN a suspicious velocity burst (e.g. 5 transactions in 2 minutes) WHEN processed by the engine THEN it receives Composite Score 0.45 <= S(x) < 0.80, generates an SMS OTP within 3 seconds, and enters PENDING_VERIFICATION state.
- GIVEN a user entering a valid 6-digit OTP within 180 seconds WHEN submitted to /api/v1/auth/verify-otp THEN the transaction transitions to APPROVED and audit event is logged.

- GIVEN an invalid OTP entered 3 consecutive times OR expired after 180 seconds WHEN evaluated THEN the transaction transitions to REJECTED/SECURITY_FLAGGED.

Sign-off: Mentor approves live prototype demo, test report, and academic documentation; all P1 critical bugs closed.

Sign-off table

Stakeholder	Role	Decision area	Signature/Approval	Date
Dr.	Project Mentor / Supervisor	Academic scope, technical methodology, final acceptance	Approved	
Keshav Pathak	AIML Project Lead	System architecture, implementation & evaluation readiness	Approved	
	External Faculty Reviewer	Capstone defense, code audit & degree evaluation	Pending Review	November 2026

3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements & Domain Research	AI/ML Lead	AI/ML Lead	Project Guide, FinTech Advisor	Capstone Committee
Data Engineering & Feature Pipeline	Data Engineer / ML Dev	AI/ML Lead	Project Guide	Team
ML Anomaly Modeling & Training	Quantitative ML Researcher	AI/ML Lead	Project Guide	Team
Backend API & Decision Engine	Backend & Systems Architect	AI/ML Lead	Cybersecurity Mentor	Team
OTP Auth & Security Implementation	Cybersecurity & Auth Engineer	AI/ML Lead	Project Guide	Team
Testing, Validation & Metrics	QA & Evaluation Analyst	AI/ML Lead	Project Guide	Committee
Final Report, Thesis & Defense	Academic Research Writer	AI/ML Lead	Project Guide	Department Head

4. Team and Roles

Member	Role	Responsibilities	Key skills	University Roll Number
Keshav Pathak	Product Lead & ML Architect	End-to-end architecture, Isolation Forest modeling, risk scoring	Python, scikit-learn, System Design	12515500221
Gaurav	Backend & Data Engineer	Feature pipeline, FastAPI REST API, database schemas & migrations	FastAPI, SQL, pandas, Docker	12515500159
Adrash Parashar	Cybersecurity & Auth Specialist	OTP verification engine, STRIDE threat model, DPDP Act compliance	Python secrets, Cryptography, OWASP	12515500036
Gauri Agnihotri	QA, UI & Documentation Lead	Monitoring dashboard, integration tests, benchmark evaluation	React, Playwright, Technical Writing	12515500161

5. Week-wise Plan and Assignments

(16-Week Academic Capstone Schedule for Semester Execution; 8 Two-Week Milestones)

Week	Milestones	Lead (ML)	Backend (Data)	Security (Auth)	QA/Docs	Deliverables	Status
1-2	Requirements & Research	Scope finalization, RBI/NPCI analysis	Data source validation	Auth threat modeling	Test strategy v0	SRS & Literature Review	Completed
3-4	Dataset Acquisition & EDA	EDA on PaySim & ULB datasets	ETL scripts & data cleaning	Mock OTP architecture	EDA test plan	Cleaned Data & EDA Report	Planned
5-6	Feature Engineering	Feature pipeline design	Sliding-window transformers	CSPRNG token service	Feature unit tests	Feature Engineering Module	Planned
7-8	ML Model Training	Isolation Forest grid search	Logistic Reg & XGBoost baselines	Attempt throttling logic	Model evaluation harness	Trained Model Artifacts	Planned
9-10	Risk Engine & Calibration	Sigmoid score blending	Decision threshold API	OTP session state machine	Boundary stress tests	Composite Risk Engine	Planned
11-12	API Backend & Dashboard	FastAPI ASGI scaffolding	DB schema & migrations	JWT/HMAC gateway security	React dashboard UI	Integrated Microservice	Planned

13-14	Hardening & Benchmarking	Latency profiling (p95<50ms)	Attack burst simulation	Replay/brute-force tests	Regression test report	Benchmark Eval Report	Planned
15-16	Release & Defense	Final sign-off & demo	Deployment packaging	Security audit review	Final thesis & slides	v1.0 Code, Thesis & Deck	Planned

6. Users and UX

6.1 Personas

Rural First-Time User Ramesh: 48, village grocery store owner in rural Maharashtra. Enters small digital transactions daily; values simplicity, safety, and clear verification prompts. Vulnerable to social engineering and confused by complex English security alerts.

Fraud Operations Analyst Anita: 29, risk operations analyst at a regional fintech payment provider. Needs real-time visibility into micro-transaction anomaly clusters, low false-positive dispute rates, and interpretable risk scores to adjust operational policies.

6.2 Top user journeys

Routine Payment Journey: Merchant scan -> Enters INR 150 -> Feature extraction -> Score S=0.18 < 0.45 -> Approved immediately without OTP. KPI: latency <50ms, zero added friction.

Anomalous Velocity Burst Journey: Attacker attempts 6th payment in 3 mins -> Feature extraction -> Score S=0.68 (Medium Risk) -> Step-Up OTP challenge triggered -> 6-digit OTP delivered via SMS -> User enters OTP -> Validated -> Approved. KPI: OTP delivery <5s, verification completed <30s.

Fraud Siphoning Attempt Journey: Attacker attempts 10 consecutive rapid transfers -> Score S=0.89 (High Risk) -> Blocked + Security Alert sent to user. KPI: Attack blocked instantly.

6.3 User stories

As a first-time rural mobile wallet user, I want my normal low-value purchases approved instantly without unnecessary OTP prompts so that my daily transactions remain fast and convenient.

GIVEN I make a regular grocery payment under normal frequency WHEN the transaction is evaluated THEN it is approved in <50ms without OTP friction.

As a vulnerable user under a phishing attack, I want the system to intercept unusual velocity bursts with an SMS OTP challenge so that my account cannot be drained without my explicit knowledge.

GIVEN rapid-fire repeated transfers occur on my account WHEN the anomaly engine flags velocity deviation THEN an SMS OTP is required before any money leaves my wallet.

6.4 Accessibility & localization

Keyboard-only navigation and screen-reader compatibility on web dashboard; contrast ratio >= 4.5:1 (WCAG AA compliant); numerical OTP display uses prominent monospace font (Consolas); responsive layout adaptable to low-cost mobile browser displays; localization support for regional vernacular prompts (Hindi, Marathi, Telugu); audio/SMS reading compatibility.

7. Market and Competitors

7.1 Competitor table

Competitor	Product	Target users	Key features	Pricing	Strengths	Weaknesses	Our differentiator
Conventional Rule Engines	Static Boolean Filters	Retail Banking	Hard threshold rules (>INR 10,000)	Low	Instant execution (<5ms)	Bypassed by micro-smurfing (<INR 500)	Sliding-window velocity & entropy profiling
FICO Falcon Fraud Manager	Enterprise Neural Profiler	Global Tier-1 Banks	Consortium analytics, neural behavioral profiling	High ($$$$)	Immense consortium database	Opaque, requires dense history, expensive	Unsupervised cold-start focus, accessible open architecture
Feedzai RiskOps	Enterprise Risk Platform	Global Fintechs & Card Networks	Ensemble ML, AutoML, Graph analytics	High ($$$$)	Real-time orchestration	High operational complexity, enterprise cost	Lightweight edge-compatible design (<50ms), specialized micro-ticket focus
Fraud Shield (Proposed)	Real-Time Micro-Txn Anomaly Engine	Rural & First-Time Mobile Wallet Users	Isolation Forest, velocity burst detection, adaptive Step-Up OTP	Open-Source / FOSS	Sub-50ms inference, cold-start population priors, zero licensing barrier	Academic prototype, requires upstream gateway integration	Dedicated micro-transaction focus, privacy-preserving tokenization

7.2 Positioning

Unique angle: Real-time behavioral anomaly detection and adaptive Step-Up authentication built specifically for low-value (<INR 1,000) digital wallet transactions in underserved rural markets.

Measurable delta: Detects low-ticket velocity attacks that 100% bypass conventional static value thresholds, while maintaining <1.0% false-positive OTP challenge rate on legitimate routine commerce.

8. Objectives and Success Metrics

O1 Latency & Throughput: p95 model inference latency <= 25 ms, total API evaluation <= 50 ms under 200 req/s load by 20 Oct 2026. KPI: p95 latency in ms.

O2 Anomaly Detection Efficacy: Precision-Recall AUC (PR-AUC) >= 0.75 on imbalanced test benchmark (<0.2% fraud), with Recall >= 75% at False Positive Rate (FPR) <= 1.0% by 25 Oct 2026. KPI: PR-AUC & Recall @ 1% FPR.

O3 Friction Minimization: Routine low-risk micro-transactions receive 0% added authentication challenges; Step-Up OTP challenge rate on legitimate population <= 1.5% by 30 Oct 2026. KPI: Legitimate Challenge Rate %.

O4 Verification Reliability: OTP token generation, SHA-256 validation, and session lifecycle execute with 100% accuracy; zero successful replay or brute-force attacks within 3-attempt limit by 05 Nov 2026. KPI: Auth security pass rate 100%.

O5 Prototype Integration & Uptime: End-to-end integration between FastAPI backend, SQLite/PostgreSQL, and React dashboard achieves >= 99.5% availability in staging environment by 10 Nov 2026. KPI: Uptime %.

9. Key Features

Feature	Description	Priority	Dependencies	Acceptance criteria
Ingestion & Validation Gateway	Ingests JSON transaction payload, validates schema via Pydantic, checks HMAC signature	Must	FastAPI, Pydantic	GIVEN valid payload WHEN submitted THEN returns 200 OK within 15ms.
Behavioral Velocity Feature Pipeline	Computes 5m/1h transaction counts, inter-arrival delta-t, recipient entropy, amount deviation ratios	Must	pandas, NumPy	GIVEN transaction stream WHEN processed THEN feature vector generated in <10ms.
Isolation Forest ML Engine	Evaluates recursive tree path lengths, outputs normalized anomaly score s(x, n) in [0, 1]	Must	scikit-learn	GIVEN feature vector WHEN scored THEN anomaly score returned in <15ms.
Composite Decision Engine	Combines ML score with rule penalties via logistic sigmoid to assign Low, Medium, or High risk tier	Must	ML Engine	GIVEN score WHEN evaluated THEN assigns correct operational tier (Allow/OTP/Block).
Step-Up OTP Authentication Service	Generates cryptographically secure 6-digit token, enforces 180s TTL, limits attempts to 3	Must	Python secrets	GIVEN Medium Risk WHEN triggered THEN OTP dispatched within 3s and verified securely.
Real-Time Monitoring Dashboard	Visualizes transaction volume, risk score distributions, alert logs, and ROC/PR performance	Should	React, Vite, Chart.js	GIVEN analyst console WHEN opened THEN displays live streaming metrics.
Vernacular Voice-Based Verification	Generates automated IVR voice OTP prompts in regional languages for non-literate users	Could	Text-to-Speech API	GIVEN audio prompt request WHEN selected THEN delivers audio code.

Detailed mathematical algorithms, state machine transitions, and edge cases are documented in Section 10 and Section 12.

10. Architecture

10.1 High-level

Clients: Mobile Wallet Simulator / Client Web App; Security Analyst Monitoring Console (React SPA).

Services: Ingestion Gateway (FastAPI ASGI); Feature Engineering Service; ML Inference Worker (Isolation Forest); Step-Up OTP Verification Service.

Data stores: Relational Database (PostgreSQL / SQLite) for transactions, assessments, and audit logs; In-memory cache for sliding-window velocity state.

Integrations: Telecom SMS Gateway (Mock / Twilio REST API); Upstream Payment Gateway Webhooks.

10.2 API spec snapshot

Endpoint	Method	Auth	Purpose	Request schema	Response schema	Codes
/api/v1/transactions/evaluate	POST	API Key	Ingest & score transaction risk	user_id, amount, recipient_vpa, timestamp, channel	200 {txn_id, risk_score, risk_tier, status, challenge_required}	200, 400, 422
/api/v1/auth/verify-otp	POST	None	Validate submitted 6-digit OTP	session_id, otp_code	200 {verified: true, txn_status: 'APPROVED'}	200, 400, 403, 410
/api/v1/auth/resend-otp	POST	None	Resend new verification token (max 2)	session_id	200 {resend: true, expires_in: 180}	200, 429
/api/v1/dashboard/metrics	GET	Bearer	Stream real-time telemetry metrics	None	200 {total_txns, anomalous_count, p95_latency, risk_distribution}	200, 401

10.3 Config and secrets

Environment variables managed via .env files (strictly Git-ignored); database credentials, API secret keys, and JWT signatures injected at runtime; cryptographic HMAC keys rotated each academic term; access to deployment secrets restricted via GitHub Actions Environment Secrets.

11. Data Design

11.1 Data dictionary

Entity	Field	Type	Null?	Allowed values	Source	Notes

User	user_id	UUID	No	RFC 4122 UUID	System	Primary Key
User	phone_hash	String(64)	No	64-char Hex	User input	Salted SHA-256 for privacy
Transaction	txn_id	UUID	No	RFC 4122 UUID	System	Primary Key
Transaction	amount	Decimal(10,2)	No	> 0.00	Wallet Client	Transaction value in INR
Transaction	recipient_vpa	String(100)	No	Valid VPA format	Wallet Client	Destination UPI handle / account
RiskAssessment	risk_score	Float	No	0.00 - 1.00	Risk Engine	Calibrated Composite Risk Score
RiskAssessment	risk_tier	Enum	No	LOW, MEDIUM, HIGH	Decision Engine	Assigned operational action
OTPSession	hashed_otp	String(64)	No	64-char Hex	OTP Service	Salted SHA-256 hashed 6-digit token

11.2 Schemas and migrations

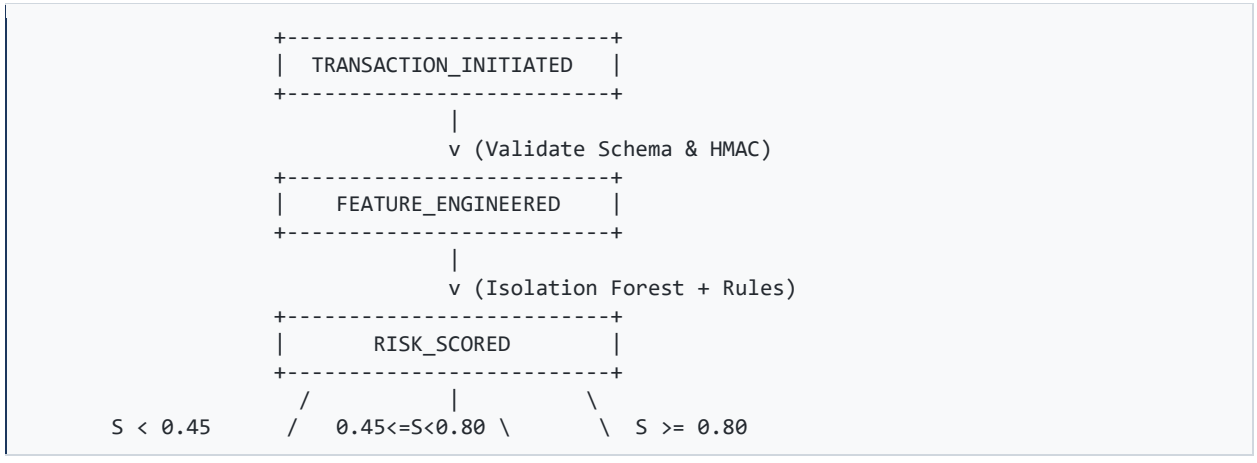
Relational schemas defined in SQLAlchemy ORM; migrations versioned using Alembic; foreign keys enforce referential integrity between Users, Transactions, Risk Assessments, and OTP Sessions; rollback scripts validated on staging.

11.3 Privacy, retention, backup/DR

PII & Retention Policy: Zero cleartext storage of phone numbers, bank account numbers, or Aadhaar credentials (DPDP Act 2023 compliance). Ephemeral OTP session states purged after 180 seconds; automated nightly database snapshots; Recovery Time Objective (RTO): 2 hours; Recovery Point Objective (RPO): 12 hours.

12. Technical Workflow Diagram

State Transition Diagram



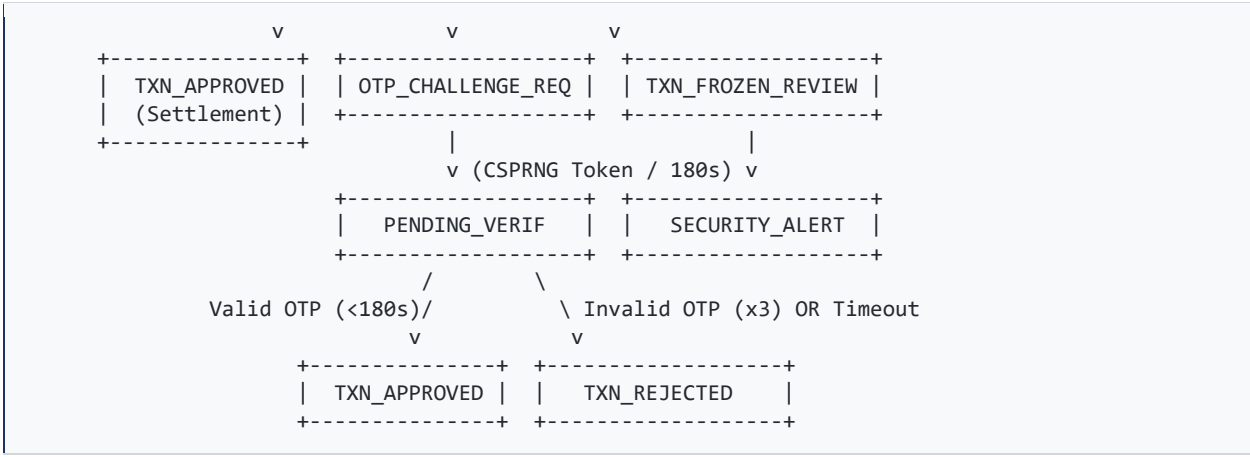


Figure 12.1: Transaction Lifecycle State Transition Diagram

Sequence Diagram

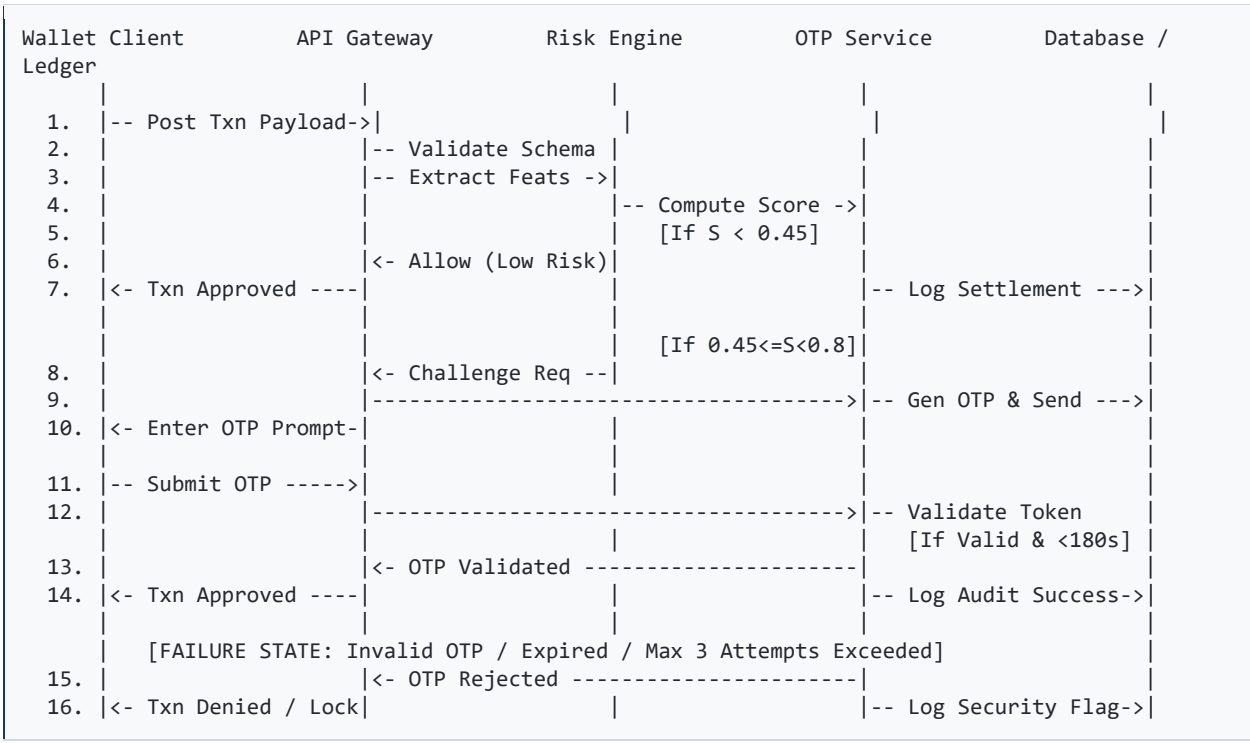
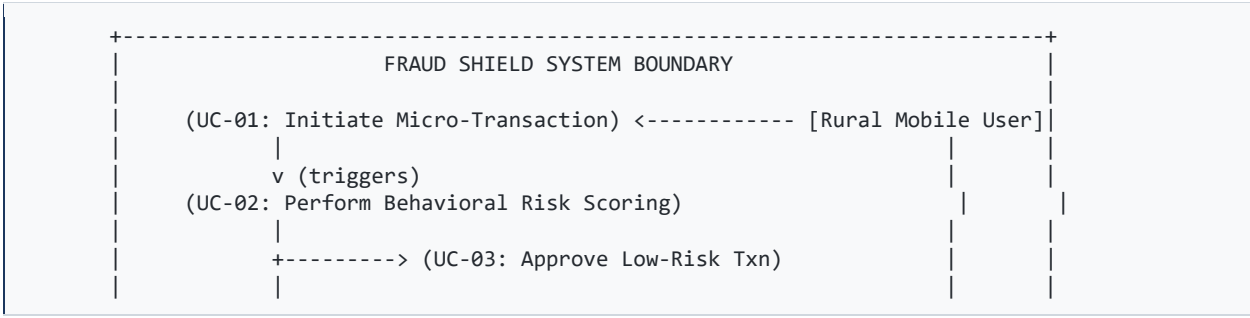


Figure 12.2: Transaction Sequence & Verification Workflow Diagram

Use Case Diagram



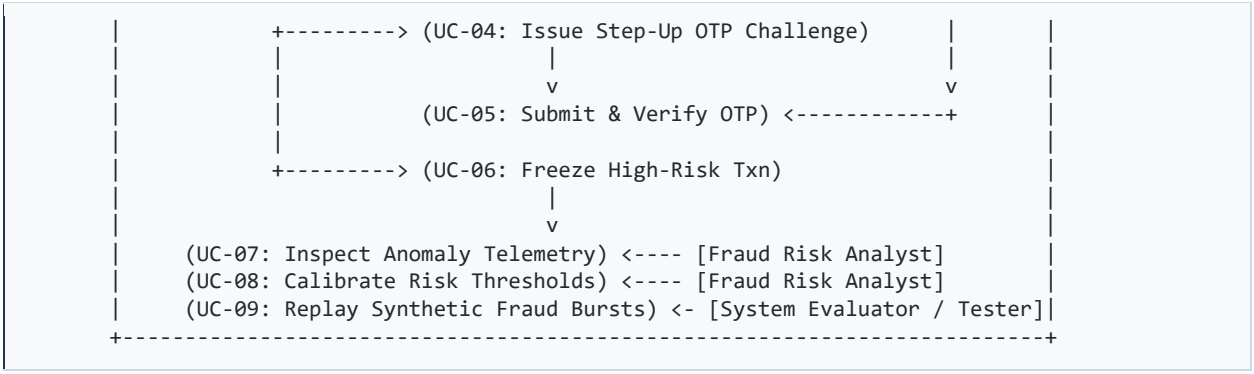


Figure 12.3: System Use Case Interaction Diagram

Data Flow Diagram

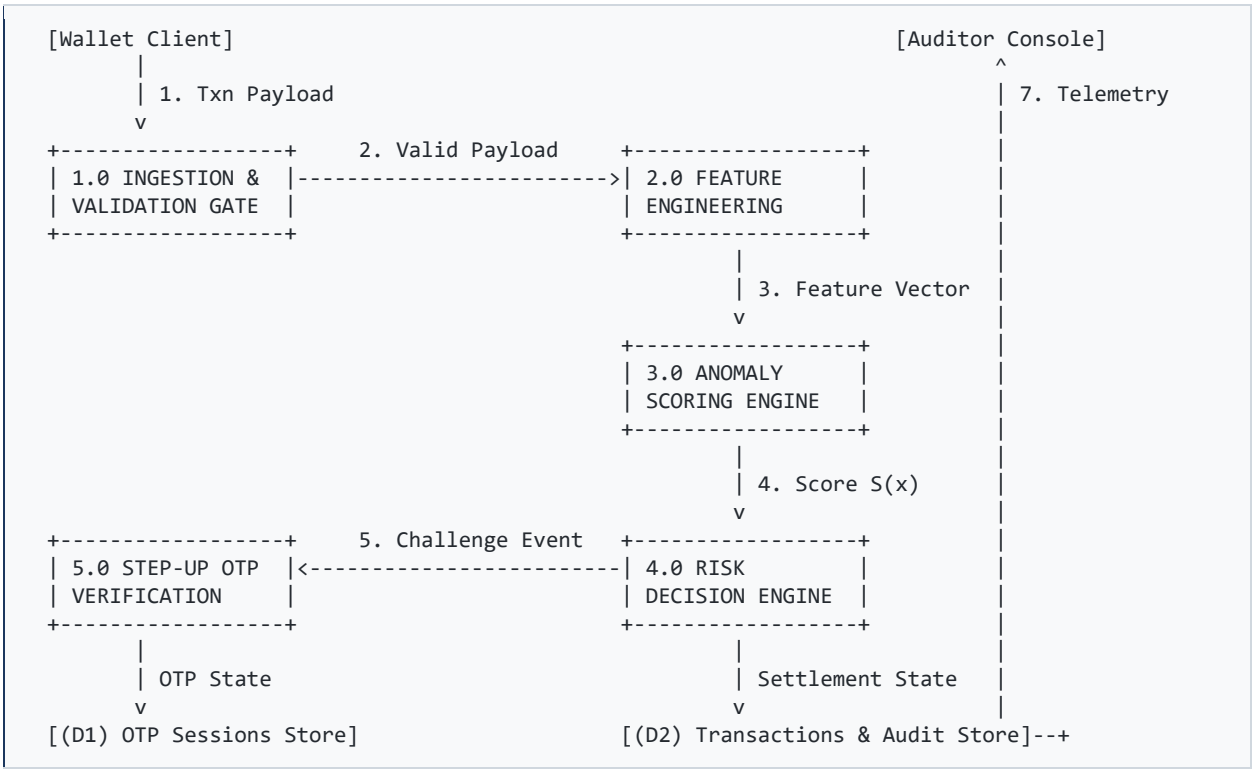
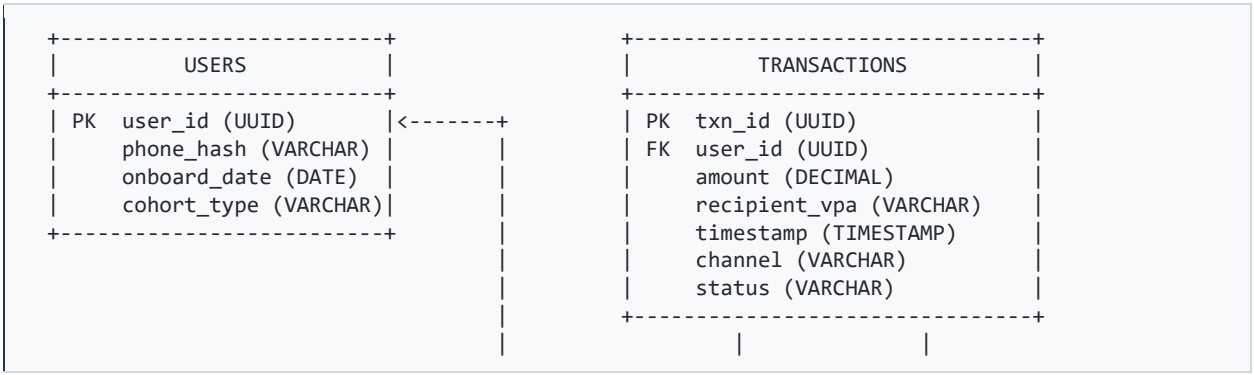


Figure 12.4: Level 1 Data Flow Diagram (DFD)

ER Diagram



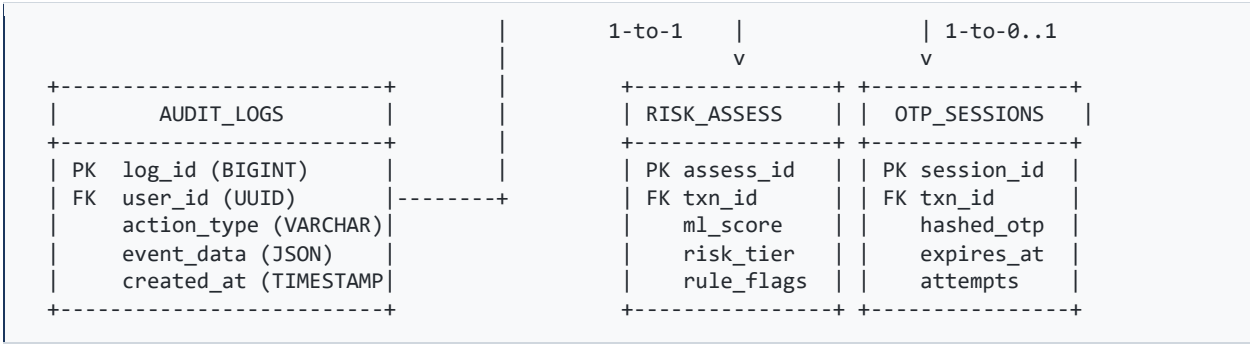


Figure 12.5: Entity-Relationship Diagram (ERD)

Technical Workflow Diagram

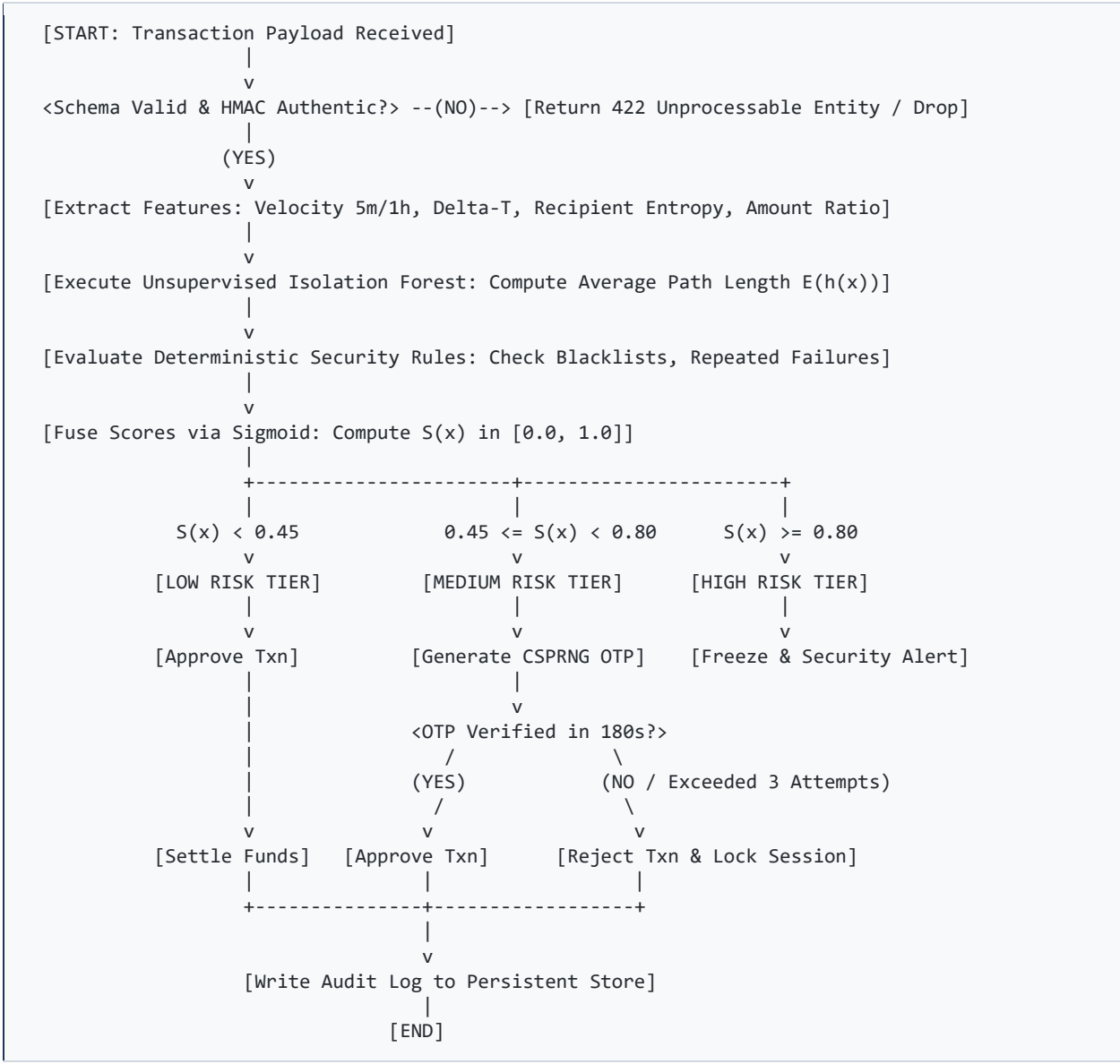


Figure 12.6: Technical Transaction Evaluation Workflow Diagram

Work Architecture Diagram



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                                                v
+-----------------------------------------------------------------------------------------------
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| PERSISTENT DATA & AUDIT STORE (PostgreSQL / SQLite)
|
|  - Transactions Ledger  - Risk Assessment Logs  - OTP Session States  - Security Audit Trail
|
+-----------------------------------------------------------------------------------------------
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```

Figure 12.7: End-to-End Component Work Architecture Diagram

13. Quality: NFRs and Testing

13.1 Non-functional requirements

Metric	SLI	Target (SLO)	Measurement
Availability	Uptime %	>= 99.5%	Automated synthetic healthcheck monitoring
Latency	p95 API response time	<= 50 ms	Application Performance Monitoring (APM)
ML Inference Latency	p95 model scoring time	<= 20 ms	In-process microsecond execution timers
Error rate	5xx HTTP response %	< 0.1%	Server access log aggregation
Security	Open critical CVEs	0	Automated Snyk & Bandit vulnerability scans

13.2 Test plan

Area	Type	Tools	Owner	Coverage target	Exit criteria
Feature & ML Engine	Unit & Validation	pytest, scikit-learn	ML Lead	>= 85%	All unit tests pass; zero NaN handling errors
API Gateway & Endpoints	Integration	pytest-asyncio, HTTPX	Backend Dev	>= 80%	All REST status codes and schemas validated
End-to-End Verification	Functional & Scenario	Playwright, Locust	QA Lead	>= 75%	Complete pass on all 3 operational risk tiers
Security & Rate Limiting	Penetration & Stress	Custom scripts, OWASP ZAP	Security Lead	100% security cases	Max 3 attempts enforced; replay blocked

13.3 Environments

Development (Local Python virtualenv, SQLite, Mock OTP) -> Staging (Docker Compose, PostgreSQL, Mock SMS Gateway, Locust load testing) -> Production Sandbox (Cloud VM / Container runtime, Managed PostgreSQL, Twilio/DLT SMS API); feature flags gate high-risk algorithmic changes.

14. Security and Compliance

14.1 Threat model (STRIDE)

Asset	Threat	STRIDE	Impact	Likelihood	Mitigation	Owner
API Gateway / Endpoints	Unauthorized transaction flood	Denial of Service	High	Medium	Token-bucket rate limiting (100 req/min), cloud firewall	Backend Dev
Verification Tokens	OTP brute-force enumeration	Spoofing / Tampering	Critical	Medium	Max 3 attempts, 180s TTL, CSPRNG token generation	Security Lead
Transaction Payloads	Man-in-the-Middle interception	Information Disclosure	High	Low	Enforce TLS 1.3 encryption, HMAC request signatures	Security Lead
User Identifiers	Phone number & account leakage	Information Disclosure	High	Low	Salted SHA-256 pseudonymization, zero PII disk storage	Security Lead
Decision Logic	Adversarial model evasion (smurfing)	Elevation of Privilege	High	Medium	Sliding-window velocity tracking across multiple time scales	ML Lead

14.2 AuthN/AuthZ

API authentication via rotating API keys and Bearer JWTs; role-based access control (Analyst, Admin) on dashboard endpoints; out-of-band Step-Up verification for challenged transactions.

14.3 Audit and logging

Immutable append-only audit trail for all risk evaluations, score calculations, and OTP validation events. Sensitive fields redacted; 90-day retention policy.

14.4 Compliance

Strictly aligned with the Reserve Bank of India (RBI) Master Directions on Digital Payment Security Controls (2021) and India's Digital Personal Data Protection (DPDP) Act 2023.

15. Delivery and Operations

15.1 Release plan

Alpha release (Week 8: Core ML models & feature engine) -> Beta release (Week 12: Integrated API, OTP service & mock simulator) -> Release Candidate (Week 14: Hardened benchmarked system) -> Final v1.0 Release (Week 16: Complete capstone release & defense).

15.2 CI/CD and rollback

GitHub Actions CI pipeline: lint (flake8/black) -> security audit (bandit) -> unit tests (pytest) -> docker build. Automated staging deployment; one-command rollback via previous Docker image tag.

15.3 Monitoring and alerting

Metric	Threshold	Alert to	Runbook
p95 API Latency	> 50 ms for 3 mins	Backend Lead	'API Latency Degradation' runbook
Error Rate (5xx)	> 1.0% over 5 mins	Tech Lead	'Error Spike & Crash Triage' runbook
OTP Failure Spike	> 15% failures in 10 mins	Security Lead	'Potential Brute-Force / SMS Outage' runbook

15.4 Runbooks

API Latency Degradation: Inspect feature cache hit rate -> check CPU/memory saturation -> profile model inference loop -> restart worker pods.

Error Spike Triage: Inspect structured application logs -> isolate failing endpoint -> verify database connectivity -> roll back recent commit if necessary.

15.5 Communication plan

Standups thrice weekly (Mon/Wed/Fri); bi-weekly sprint reviews and progress demos with Project Guide; shared Git issues and design documentation.

16. Risks and Mitigations

16.1 Risk heatmap

Risk	Probability	Impact	Score	Mitigation	Owner	Status
Dataset domain divergence (PaySim vs. Indian UPI)	Medium	High	12	Augment with synthetic micro-velocity bursts and log-transformed amounts	ML Lead	Open
Real-time inference latency exceeding 50ms SLA	Medium	High	12	Pre-compute rolling user velocity baselines, limit tree depth (n_samples=256)	Backend Dev	Open
SMS gateway latency or delivery failure	Medium	Medium	9	Decoupled mock SMS driver for testing, fallback resend logic with backoff	Security Lead	Open

Extreme class imbalance degrading PR-AUC	High	Medium	12	Use Isolation Forest unsupervised anomaly isolation + SMOTE on supervised benchmarks	ML Lead	Open

17. Research and Evaluation

Market review: Comprehensive review of Indian digital payment statistics from RBI Annual Reports (131B+ UPI transactions in FY24), NPCI monthly bulletins, CERT-In advisories, and comparative analysis of enterprise solutions (FICO Falcon, Feedzai, Featurespace).

Evaluation plan: Quantitative benchmarking on PaySim (6.3M records) and ULB Credit Card Fraud (284k records) datasets; rigorous measurement of Precision, Recall, PR-AUC, False Positive Rate at 95% Recall, and p95 inference latency across candidate models.

Limitations: The academic prototype operates on historical and simulated transaction streams; live deployment within commercial bank cores requires authorized production sandbox access and DLT telecom registration.

18. Appendices

Glossary: UPI (Unified Payments Interface), VPA (Virtual Payment Address), Isolation Forest, PR-AUC (Precision-Recall Area Under Curve), OTP (One-Time Password), CSPRNG (Cryptographically Secure Pseudo-Random Number Generator), DPDP Act (Digital Personal Data Protection Act, 2023), SLI/SLO (Service Level Indicator / Objective), RTO/RPO (Recovery Time / Point Objective).

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